

Loop Interface Exchanger

A shell and tube design, for heating the Helisol 5A shell side, using the heat from the Loop 2 ternary eutectic mixture transfer fluid, tube side. Base equations taken from Heat Exchanger Design Coursework from semester 1.

```
> restart;
> #Equations
R := (T_hot1 - T_hot2)/(T_cold2 - T_cold1);
R_gas := 8.314;
S := (T_cold2 - T_cold1)/(T_hot1 - T_cold1);
A_req := Q/(F*U*DeltaT_LM);
DeltaT_LM := ((T_hot2 - T_cold1)-(T_hot1 - T_cold2))/ln(
(T_hot2-T_cold1)/(T_hot1-T_cold2));
T_hot2 := T_hot1 - Q/(m_hot*c_hot); #Outlet temperature of
hot fluid
```

$$\begin{aligned}
 R &:= \frac{T_{hot1} - T_{hot2}}{T_{cold2} - T_{cold1}} \\
 R_{gas} &:= 8.314 \\
 S &:= \frac{T_{cold2} - T_{cold1}}{T_{hot1} - T_{cold1}} \\
 A_{req} &:= \frac{Q}{F U \Delta T_{LM}} \\
 \Delta T_{LM} &:= \frac{T_{hot2} - T_{cold1} - T_{hot1} + T_{cold2}}{\ln\left(\frac{T_{hot2} - T_{cold1}}{T_{hot1} - T_{cold2}}\right)} \\
 T_{hot2} &:= T_{hot1} - \frac{Q}{m_{hot} c_{hot}}
 \end{aligned} \tag{1.1}$$

```
> #1 is in
#2 is out
#Cold is Helisol 5A
#Hot is ternary eutectic mixture

print("====Specifications====");
```

```

m__cold := 0.25;#kg/s      (Helisol 5A mass) (Loop 1 fluid)
T__cold2 := 310 + 273.15;#Kelvin (Helisol 5A Temp required for
air heater)
m__hot := 0.7609;#kg/s      (ternary eutectic mixture mass) (Loop
2 fluid)

T__cold1 := 263.0483336 + 273.15; #Kelvin (Helisol 5A From Tor
Heating Jacket)
Q := m__cold*c__cold*(T__cold2 - T__cold1); #W

T__hot1 := 620 + 273.15; #Kelvin
c__hot := 2300; #J/kg/K (Isobaric) [Same as cp]
c__cold := 2150; #J/kg/K (Isobaric) [Same as cp]

print ("====Properties of ternary eutectic mixture, tube side====
");
rho__t := 2050; #kg/m3
mu__t := 0.0852*exp((3.51e4)/(R__gas*T__hot1))/1000; #Pa*s;
k__t := 0.55; #W/m/K
C__pt := 2300; #J/kg/K

print("=====Estimated=====");
U__o__ass := 83.165; #W/m2/K

```

"====Specifications===="

$$m_{cold} := 0.25$$

$$T_{cold2} := 583.15$$

$$m_{hot} := 0.7609$$

$$T_{cold1} := 536.1983336$$

$$Q := 11.73791660 \, c_{cold}$$

$$T_{hot1} := 893.15$$

$$c_{hot} := 2300$$

$$c_{cold} := 2150$$

"====Properties of ternary eutectic mixture, tube side===="

$$\rho_t := 2050$$

$$\mu_t := 0.009622514932$$

$$k_t := 0.55$$

$$C_{pt} := 2300$$

$$\text{"====Estimated===="}$$

$$U_{o_ass} := 83.165$$

(1.2)

```
> R := R;
  S := S;
  A_req;
  T_hot2;
  T_cold2;
```

$$R := 0.3071305729$$

$$S := 0.1315350811$$

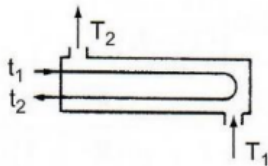
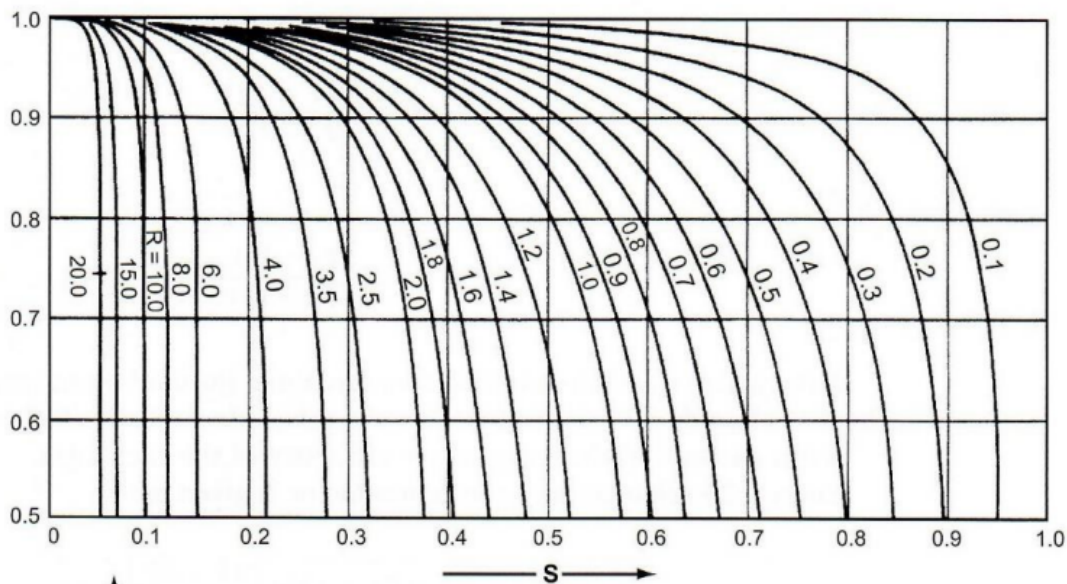
$$\frac{77.41377923}{FU}$$

$$878.7297078$$

$$583.15$$

(1.3)

The "correction factor" F



Temperature correction factor: one shell pass;
two or more even tube passes.

```
> F := 0.95; #VERY ROUGH, NEED TO MEASURE PROPERLY
      F := 0.95
```

(1.4)

```
> U := U_o_ass; #TEMPORARY ASSUMPTION
  A_req; #m2
  DeltaT_LM;
```

$$U := 83.165$$

$$0.9798375358$$

$$325.9952032$$

(1.5)

```
> L__tube := 0.7; #m
D__internal := 0.026; #m
D__external := 0.030; #m
#Plain tubes (no fins)
A__o_one_tube := Pi*D__external*L__tube;
N__tt := A__req/A__o_one_tube;
```

$$L_{tube} := 0.7$$

$$D_{internal} := 0.026$$

$$D_{external} := 0.030$$

$$A_{o_one_tube} := 0.06597344573$$

$$N_{tt} := 14.85199878$$

(1.6)

```
> #Chosen Layout is a triangular with 30 degrees rotation.
#Number of passes is 2
N__p := 2;
psi__n := 0.17;
C__1 := 0.866;
L__tp := 1.25 * D__external; #Minimum TEMA standars is 1.25
https://www.thermopedia.com/content/1121/
```

$$N_p := 2$$

$$\psi_n := 0.17$$

$$C_l := 0.866$$

$$L_{tp} := 0.03750$$

(1.7)

```
>
D__ctl := L__tp*(4*C__1*N__tt/3.14159265/(1-psi__n))^(1/2);
D__ctl := 0.1665707264
```

(1.8)

```
> L__bb := 0.0127; #Baffle bypass clearance
L__bb := 0.0127
```

(1.9)

```
>
D__s := D__ctl + L__bb + D__external;
L__B := 0.5*D__s; #Baffle spacing
D__s := 0.2092707264
```

(1.10)

$$L_B := 0.1046353632 \quad (1.10)$$

```
> P__v := L__tp * cos(3.14159265/6);
N__r := D__ctl / P__v;
BaffleCut := 0.25; #%
N__tcc := (1-BaffleCut)*N__r;
```

$$P_v := 0.03247595265$$

$$N_r := 5.129048197$$

$$BaffleCut := 0.25$$

$$N_{tcc} := 3.846786148 \quad (1.11)$$

```
> T__coldAvg := (T__cold1 + T__cold2)/2;
```

$$T_{coldAvg} := 559.6741668 \quad (1.12)$$

```
> #Calculating Tube side cross-sectional flow area
N__per_pass := N__tt/N__p;
A__internal := (3.14159/4)*D__internal^2; #m2
A__t := N__per_pass * A__internal; #m2
```

$$N_{per_pass} := 7.425999390$$

$$A_{internal} := 0.0005309287100$$

$$A_t := 0.003942676277 \quad (1.13)$$

```
> #Helisol 5A key equations
v__t := m__hot/(rho__t*A__t);
Re__t := (rho__t*v__t*D__internal)/mu__t;
Pr__t := (C__pt*mu__t)/k__t;
mu__corr := 1;
Nu__t := 1.86*(Re__t*Pr__t*D__internal/L__tube)^(1/3);
h__i := (Nu__t *k__t)/D__internal; #W/m^2/K
```

$$v_t := 0.09414182287$$

$$\Re_t := 521.4602622$$

$$Pr_t := 40.23960789$$

$$\mu_{corr} := 1$$

$$N_t := 17.11708076$$

$$h_i := 362.0920930 \quad (1.14)$$

```
> #properties of Helisol 5A, shell side
rho__s := 689.5; #kg/m3 (air at 280C)
mu__s := 0.41e-3; #kg/m/s
k__s := 0.076; #W/m/K
C__ps := 2150;
```

$$\rho_s := 689.5$$

$$\mu_s := 0.00041$$

$$k_s := 0.076$$

$$C_{ps} := 2150$$

(1.15)

```
> Q_vol_s := m_cold/rho_s;
A_face := L_tube*(D_ctl + D_external);
u_f := Q_vol_s/A_face;
u_max := u_f*(L_tp/(L_tp - D_external));
```

$$Q_{vol_s} := 0.0003625815809$$

$$A_{face} := 0.1375995085$$

$$u_f := 0.002635049971$$

$$u_{max} := 0.01317524985$$

(1.16)

```
> Pr_s := C_ps*mu_s/k_s;
Re_s := rho_s * u_max * D_external/mu_s;
Nu_s := 0.27*Re_s^0.63*Pr_s^0.36;
h_s := Nu_s * k_s/D_external;
```

$$Pr_s := 11.59868421$$

$$Re_s := 664.7074224$$

$$Nu_s := 39.15688519$$

$$h_s := 99.19744247$$

(1.17)

```
> #Defining Thermal Conductivity Variables
k_tube := 50; #W/m/K
Rf_o := 0.0002; #m2*K/W
Rf_i := 0.0003; #m2*K/W
```

$$k_{tube} := 50$$

$$Rf_o := 0.0002$$

$$Rf_i := 0.0003$$

(1.18)

```
> #Outside area calculations
```

```
A_o := A_o_one_tube * N_tt;
```

$$A_o := 0.9798375355$$

(1.19)

```
> #Internal Area calculations
```

```
A_i := 3.14159 * D_internal * L_tube * N_tt;
```

$$A_i := 0.8491918134$$

(1.20)

```
> eta_o := 1.0; #No fin efficiency losses
```

$$\eta_o := 1.0$$

(1.21)

```

> #Calculating all thermal resistances
R__shell_conv := 1/eta__o/h__s; #Air side convection
R__tube_conv := 1/(h__i*(A__i/A__o)); #Water side convection
R__wall_cond := (D__external * ln(D__external/D__internal))/(2*
k__tube);
R__fouling := (Rf__i * (A__o/A__i)) + Rf__o;


$$R_{shell\_conv} := 0.01008090506$$


$$R_{tube\_conv} := 0.003186612331$$


$$R_{wall\_cond} := 0.00004293025314$$


$$R_{fouling} := 0.0005461541385 \quad (1.22)$$


```

```

> #Calculating U
R__total := R__shell_conv + R__tube_conv + R__wall_cond +
R__fouling;


$$R_{total} := 0.01385660178 \quad (1.23)$$


```

```

> print("=====");
U__o_calc := 1/R__total; #W/m2/K
print("=====");

"====="

$$U_{o\_calc} := 72.16776637$$

"=====" \quad (1.24)

```

```

> #Tube side pressure drop

f__t := 16/Re__t;
L__total := L__tube * N__p;
K := 1.8*N__p;
DP__t_friction := 4*f__t*(L__total/D__internal)*(rho__t*v__t^2/2)
;
DP__t_return := K*(rho__t*v__t^2)/2;
DP__t := DP__t_friction + DP__t_return;


$$f_t := 0.03068306669$$


$$L_{total} := 1.4$$


$$K := 3.6$$


$$DP_{t\_friction} := 60.03472354$$


$$DP_{t\_return} := 32.70329958$$


$$DP_t := 92.73802312 \quad (1.25)$$


```

```

> #Shell side pressure drop
A__s := (L__tp - D__external)*D__s*L__B/L__tp;

```

```

G__s := m__cold/A__s;
u__s := G__s/rho__s;
d__e := 1.10/D__external*(L__tp^2 - 0.917*D__external^2);
Re__s := G__s*d__e/mu__s;

```

$$A_s := 0.004379423693$$

$$G_s := 57.08513666$$

$$u_s := 0.08279207637$$

$$d_e := 0.02130150000$$

$$\Re_s := 2965.851315$$

(1.26)

```
> #Calculated via Robinson & Briggs
```

```

f := exp(0.576 - 0.19*ln(Re__s));
DP__s := N__r * f * (rho__s * u__max^2 / 2);
f := 0.3894442002

```

$$DP_s := 0.1195374576$$

(1.27)

```
> print("====summary====");
```

```

Re__t := Re__t;
Re__s := Re__s;
DP__t := DP__t; #Pa
DP__s := DP__s; #Pa
U__o_calc := U__o_calc;
U__o_ass := U__o_ass;
Q := Q/1000; #kW

```

"====summary===="

$$\Re_t := 521.4602622$$

$$\Re_s := 2965.851315$$

$$DP_t := 92.73802312$$

$$DP_s := 0.1195374576$$

$$U_{o_calc} := 72.16776637$$

$$U_{o_ass} := 83.165$$

$$Q := 25.23652069$$

(1.28)

